Semantic-Guided Imitation Learning with History-Conditioned Feedback

***Abstract*—**

***Index Terms*—Imitation learning, semantic representations, visuomotor control, residual feedback.**

I. Introduction

Reliable execution of sequential manipulation tasks is essential for practical robotic applications but remains challenging [1]. Sequential manipulation tasks require a robot to achieve an overall goal through a sequence of interrelated operations [2]. Such tasks arise in practical applications ranging from household cleaning, where a tablecloth is lifted to clear debris beneath it and then restored, to surgery, where tissue is retracted to expose the target region before resection [3]. A key challenge of these tasks is that preceding interactions change the conditions under which subsequent operations can be performed. For example, compared with simple pick-and-place tasks involving visible and accessible targets, manipulation under occlusion requires the robot to assess whether prior actions have made the target visible and accessible or further manipulation is needed [4], [5]. Consequently, reliable execution requires a policy that adapts subsequent actions to interaction outcomes observed through visual feedback.

Visuomotor imitation learning offers a practical approach to sequential manipulation by reducing the need for explicit interaction modeling, but reliable execution remains difficult with limited demonstrations. It learns a policy that maps visual and proprioceptive observations to actions using expert demonstrations. For example, Action Chunking with Transformers (ACT) [6] has been used in Mobile ALOHA to learn coordinated mobile manipulation, including opening a cabinet and storing a cooking pot [7]. Similarly, Diffusion Policy [8] has been adopted in the Universal Manipulation Interface (UMI) framework to learn bimanual cloth folding and multi-step dish washing [9]. However, execution errors can lead a policy into states poorly covered by limited demonstrations, where its predicted actions may no longer suit the actual task conditions, compromising subsequent operations and task completion [10].

Prior work has focused on improving the robustness of visuomotor policies to execution deviations by expanding data coverage, improving visual representations, and adapting actions during execution. Ross et al. [10] expanded the training dataset with expert action labels for states visited by the learned policy, while Laskey et al. [11] injected noise into expert actions during demonstration collection to obtain examples of recovery behavior. Although data coverage expansion improves coverage of states arising from execution deviations, it requires additional expert supervision or modifications to demonstration collection, limiting its applicability when only a fixed demonstration set is available. Zhu et al. introduced object-centric visual representations for robust visuomotor imitation [12] and incorporated 3D object representations to improve generalization across backgrounds, viewpoints, and object instances [13]. Although these methods improve generalization to visual variations, robust object representations alone do not ensure that subsequent actions remain appropriate when execution deviates from demonstrations. Ankile et al. [14] trained a residual policy through reinforcement learning to correct the actions of a frozen imitation policy, while Sendai et al. [15] learned corrections from demonstrations using updated observations and base-policy context to refine actions during chunk execution. However, reinforcement-based correction requires additional environment interaction, whereas corrections based on updated observations and base-policy context may be insufficient when appropriate adjustments depend on preceding actions and their observed effects.

In this work, we develop a theory-guided visuomotor imitation learning framework that combines task-relevant representation learning with history-conditioned feedback to support sequential manipulation from limited demonstrations. Our analysis derives conditions linking imitation error and representation quality to task-failure bounds. It further identifies when updated observations and execution history provide additional information for predicting residual action errors, and when learned corrections can realize this benefit. Guided by this analysis, we supervise policy query tokens with task descriptors derived from semantic masks, encouraging the policy to retain information relevant to changing manipulation conditions. We then train a residual model that combines updated observations, the supervised query features, and action–response histories to adjust pending actions in the current plan according to observed interaction outcomes. Both the base policy and the residual model are trained from the same fixed demonstration set, without additional robot interaction for policy training. Simulation and real-world experiments demonstrate the effectiveness of the proposed framework in sequential manipulation tasks.

II. Related Work

III. Problem Formulation and Theoretical Analysis

This section studies conditions for improving task performance from limited demonstrations. Starting from the relationship between imitation error and task success, we examine two paths for improvement: preserving information needed to distinguish the task consequences of different actions, and using new observations and execution history to produce effective corrections.

We first establish conditions under which action error bounds task-failure probability and explain why improved coverage in representation space alone does not guarantee better task performance. We then examine how semantic guidance can preserve task-relevant information and, under consistent expert selection, control imitation-error transfer through expert-action reconstruction. Finally, we characterize the predictive value of updated observations and history, and identify when learned residual corrections reduce imitation error and tighten the task-performance bound.

A. Visuomotor Imitation Learning

We study offline visuomotor imitation learning from a finite, fixed set of robot demonstrations, for manipulation tasks whose completion can be assessed from visual history. The policy generates control commands from visual and proprioceptive observations and available execution history. Training is based on this fixed demonstration dataset, with demonstrated actions providing the primary imitation target, whereas deployment performance is measured by the probability of task completion over a finite horizon.

1) System and Observations: Consider a robotic system governed by a single decision policy that performs a given manipulation task through the joint actions of its actuators. The joint robot-environment state $s_t \in \mathcal{S}$ evolves under the command $u_t \in \mathcal{A}$ and gives rise to observation o_t. The observation $o_t = (I_t^{1:V}, q_t)$ comprises multi-view images $I_t^{1:V} = (I_t^1, \dots, I_t^V)$ and proprioceptive information q_t, where $V \geq 1$ is the number of cameras and I_t^v is the image from the vth view. Over a finite horizon of T control steps, this interaction is represented by the following partially observed controlled process:

$$\begin{aligned} s_0 &\sim p_0, \\ s_{t+1} &\sim P(\cdot \mid s_t, u_t), \quad 0 \leq t < T, \\ o_t &\sim O(\cdot \mid s_t), \quad 0 \leq t \leq T. \end{aligned} \tag{1}$$

Here, $\mathcal{S}$ is the state space, $\mathcal{A} \subseteq \mathbb{R}^{d_u}$ is the admissible command set specified by the chosen robot control interface, d_u is the command dimension, p_0 is the initial-state distribution, and P and O are the transition and observation kernels, respectively.

2) Offline Demonstrations and Policy: The policy learns from a fixed demonstration set $\mathcal{D} = \{\zeta_i\}_{i=1}^N$ for the same manipulation task. The ith demonstration trajectory ζ_i records observations and demonstrated control commands:

$$\zeta_i = (o_0^i, u_0^{\mathrm{E},i}, o_1^i, \dots, u_{T_i-1}^{\mathrm{E},i}, o_{T_i}^i). \tag{2}$$

Here, N is the finite number of demonstrations, $T_i \geq 1$ is the number of control steps in the ith trajectory, and the superscript E denotes demonstrated commands.

The information available to the deployed policy π is represented by the history $\mathcal{H}_t$. This history includes observations $o_{0:t}$ obtained up to time t, previously executed commands $u_{0:t-1}$, and other runtime records before the current decision. The policy defines a distribution over admissible control commands, with $u_t \sim \pi(\cdot \mid \mathcal{H}_t)$. The command history is empty at $t = 0$.

3) Learning and Task Objectives: For the following analysis, we consider deterministic policies acting on the available history, with π_{E} denoting the expert policy. Let ν denote the population distribution underlying the demonstrations, where $(t, h) \sim \nu$ specifies a control step and its available history. We define the population imitation risk as the expected squared Euclidean deviation from the expert's action:

$$R_\nu(\pi) := \mathbb{E}_{(t,h)\sim\nu} \|\pi(h) - \pi_{\mathrm{E}}(h)\|^2. \tag{3}$$

Training seeks to reduce this risk through an empirical imitation loss on the finite demonstrations; the specific training losses are defined in Section IV.

The deployment objective is to complete the task within the finite horizon. The measurable task-completion function $\Phi : \mathcal{S}^{T+1} \to \{0, 1\}$ equals 1 for state trajectories satisfying the task's terminal or temporal requirements and 0 otherwise. This defines the completion indicator Ψ and the task-completion probability $J(\pi)$ of the deployed policy π:

$$\Psi = \Phi(s_{0:T}), \qquad J(\pi) = \mathbb{P}_\pi(\Psi = 1). \tag{4}$$

Here, $\mathbb{P}_\pi$ is the trajectory probability law induced jointly by the system model and the complete deployed policy π. Learning aims to reduce $R_\nu(\pi)$, whereas deployment aims to increase $J(\pi)$. A lower imitation risk does not by itself guarantee a higher task-completion probability; the following analysis examines conditions relating these objectives.

4) Visual Assessability: Under a fixed camera configuration and evaluation horizon T, the evaluation domain $\Omega_{\text{eval}} \subseteq \mathcal{S}^{T+1} \times \mathcal{I}^{T+1}$ consists of prespecified pairs of physical trajectories and visual histories consistent with the system model and visibility conditions, where $\mathcal{I}$ is the space of multi-view image tuples $I_t^{1:V}$.

Assumption 1 (Visual assessability): There exists a measurable visual decision function $\mathcal{V} : \mathcal{I}^{T+1} \to \{0, 1\}$ such that, for every $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$,

$$\Phi(s_{0:T}) = \mathcal{V}(I_{0:T}^{1:V}). \tag{5}$$

Assumption 1 specifies visual identifiability of completion within the evaluation domain, without requiring full-state observability. The subsequent analysis considers policy executions and counterfactual action comparisons for which $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$ almost surely.

B. From Imitation Error to Task Performance

Whether action error bounds task failure depends on where deviations occur and their subsequent task consequences. We establish this connection and show that transferring imitation risk from demonstrations to deployment depends jointly on representation-level coverage and conditional action-error discrepancy.

1) Task-Performance Difference Between Policies: We compare a policy π with a reference policy π_0 through their task-failure probabilities and express the difference as a sum of expected single-step action consequences. Each consequence is defined by comparing alternative actions at the same history while following π_0 afterward [16], [17]. This decomposition

provides the basis for relating imitation error to task performance.

We consider deterministic policies acting on the available history. Given a history $\mathcal{H}_t = h$ and a candidate action $a \in \mathcal{A}$, define the action-value function as the probability of task failure ($\Psi = 0$):

$$Q_t^{\pi_0}(h, a) := \mathbb{P}^{a,\pi_0}(\Psi = 0 \mid \mathcal{H}_t = h). \tag{6}$$

Here, $\mathbb{P}^{a,\pi_0}$ denotes the probability law obtained by taking action a at time t and following π_0 afterward.

When the current action is also selected by π_0, this reduces to the value function of π_0, $V_t^{\pi_0}(h) := Q_t^{\pi_0}\big(h, \pi_0(h)\big)$. The effect of choosing a instead of $\pi_0(h)$ is therefore measured by

$$A_t^{\pi_0}(h, a) := Q_t^{\pi_0}(h, a) - V_t^{\pi_0}(h). \tag{7}$$

Thus, $A_t^{\pi_0}(h, a) > 0$ indicates an increase in failure probability, whereas $A_t^{\pi_0}(h, a) < 0$ indicates a decrease.

To relate these action-level consequences to differences in task performance between policies, we require the following common comparison conditions.

Assumption 2 (Common task and environment): The compared policies share the same initial distribution and task-completion criterion. Given a history h and an action a, the next available history follows a conditional distribution that does not otherwise depend on the policy. The reference policy π_0 is defined on all histories involved in the comparison.

Let $F(\pi) = 1 - J(\pi)$ denote the task-failure probability of policy π. To evaluate π relative to π_0, we set $a = \pi(h)$ and aggregate the resulting action consequences along trajectories generated by π.

Applying the finite-horizon performance-difference identity [16]–[18] to these one-step action effects gives

$$F(\pi) - F(\pi_0) = \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))\right]. \tag{8}$$

Here, $\mathbb{E}_\pi$ denotes expectation over trajectories generated by π. Equation (8) evaluates action effects at the situations actually visited by the evaluated policy π. Its proof is given in Appendix A-A. This relation allows task-performance differences to be bounded through the consequences of action deviations.

2) Task-Failure Bounds from Imitation Risk: Taking the expert policy as the reference, we show that the imitation risk $R_\nu(\pi)$ bounds task-failure probability $F(\pi)$ when demonstrations sufficiently cover deployment histories and action deviations have bounded task consequences.

Imitation risk is evaluated under the demonstration distribution ν, but deployment histories may follow a different distribution because actions depend on π. Let μ_π^t denote the distribution of $(t, \mathcal{H}_t)$ at control step t under π. The distribution $\bar{\mu}_\pi = T^{-1}\sum_{t=0}^{T-1}\mu_\pi^t$ corresponds to choosing a control step uniformly and observing its history.

Assumption 3 (Demonstration coverage):

For the evaluated policy π, there exists a uniform constant $1 \le C < \infty$ such that

$$\bar{\mu}_\pi(\mathcal{B}) \le C\,\nu(\mathcal{B})$$

for every measurable set $\mathcal{B}$ of pairs (t, h).

This condition excludes deployment situations absent from the reference distribution and limits how much their probabilities can be underestimated.

Assumption 4 (Action sensitivity): Let A_t^{E} denote the action effect with the expert policy as the reference. For a fixed action scale, there exists a uniform constant $0 \le L < \infty$ such that

$$|A_t^{\mathrm{E}}(h, a)| \le L\|a - \pi_{\mathrm{E}}(h)\|$$

for all relevant (t, h, a).
This assumption allows action deviations to be translated into bounds on changes in task-failure probability.

Following the connection between deployment imitation error and task cost [10], Assumptions 3 and 4, together with the performance-difference identity, yield the following bound (Appendix A-B):

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{CR_\nu(\pi)}. \tag{9}$$

Here, C accounts for the shift from demonstration to deployment distributions, while L measures the sensitivity of task outcomes to action deviations. With other quantities fixed, a smaller $R_\nu(\pi)$ yields a tighter upper bound on task-failure probability.

This bound identifies coverage and action sensitivity as key conditions for translating imitation accuracy into task-performance guarantees. We next examine the limitations of complete-history coverage and the role of representations in risk transfer.

3) Imitation-Risk Transfer Through Representations: To address the limitations of complete-history coverage, we analyze imitation-risk transfer in representation space. We derive a bound showing that compression can relax coverage requirements, while differences in conditional imitation error must also be accounted for.

Assumption 3 can fail even after an arbitrarily small action deviation. For $T \ge 2$, suppose the expert always executes $u_0 = 0$, whereas the learned policy executes an admissible $u_0 = \varepsilon \ne 0$. Histories recording $u_0 = \varepsilon$ then occur under the learned policy but have zero probability under the expert demonstration distribution. No finite coverage constant C can satisfy the assumption.

A representation can merge such distinctions, potentially relaxing coverage requirements, but may also discard information needed to distinguish action consequences. This tension between distributional alignment and information loss under a noninvertible representation also appears in domain-adaptation analyses [19], [20].

For a fixed policy, expert, and common measurable causal encoder $Z = \phi(t, \mathcal{H}_t)$ retaining the time index, let μ_Z and ν_Z be the representation distributions induced by $\bar{\mu}_\pi$ and ν. In place of Assumption 3, suppose $\mu_Z(\mathcal{B}) \le C_\phi \nu_Z(\mathcal{B})$ for every measurable representation event $\mathcal{B}$, with $1 \le C_\phi < \infty$. Even at the same representation, the two distributions may contain different histories. Let $e(h) = \|\pi(h) - \pi_{\mathrm{E}}(h)\|$ and define their conditional-error discrepancy as $D_\phi = \mathbb{E}_{z\sim\mu_Z}|m_\mu(z) - m_\nu(z)|$, where $m_\mu(z) = \mathbb{E}_{\bar{\mu}_\pi}[e(h)^2 \mid Z = z]$ and $m_\nu(z) = \mathbb{E}_\nu[e(h)^2 \mid Z = z]$. With square-integrable action error under

both distributions and measurable versions of these conditional means,

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \le C_\phi R_\nu(\pi) + D_\phi. \tag{10}$$

If Assumption 4 also holds, the performance-difference identity gives

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C_\phi R_\nu(\pi) + D_\phi}. \tag{11}$$

Appendix A-C provides the proof. With imitation risk and action sensitivity fixed, (11) shows that the effect of a representation on the bound depends jointly on the coverage factor C_ϕ and the conditional error discrepancy D_ϕ. This motivates seeking representations that reduce C_ϕ while controlling D_ϕ, so that their combined contribution $C_\phi R_\nu(\pi) + D_\phi$ decreases.

C. Task-Relevant Information Preservation

We seek conditions under which a representation can improve coverage while controlling the conditional imitation-error discrepancy. We examine semantic guidance as a way to retain task-relevant information, then connect the accuracy of that information to expert-action reconstruction and a bound on D_ϕ.

1) Semantic-Guided Compression and Coverage: Semantic guidance can encourage an encoder to merge appearance differences while retaining object-level structure relevant to action selection. This motivates a representation design, not an automatic coverage guarantee. As in Section III-B3, compression can relax coverage requirements by making some history distinctions irrelevant to representation events. Appending a deterministic semantic map while retaining the original RGB input does not itself change the best coverage constant relative to RGB, since projection recovers that input. The relevant object is the learned decision representation $Z = \phi(t, \mathcal{H}_t)$, not the number of input modalities.

Improved coverage alone does not ensure a tighter bound; the error associated with merging histories must also be accounted for. A sufficient route is to retain information from which a common rule can reconstruct expert actions under both demonstration and deployment distributions. This emphasis on decision-relevant information is consistent with information-state and behaviorally grounded representation analyses [21], [22]. We next formulate this requirement using object attributes and spatial relationships.

2) Task Information and Expert-Action Reconstruction: We examine when retained task information is sufficient to approximate expert choices. The reconstruction error depends on both what the selected information fails to explain and how accurately it is estimated.

Fix the evaluated policy π and set $\mu = \bar{\mu}_\pi$, with ν the demonstration distribution. Write $\pi(h) = p(Z)$ for the same measurable policy readout under both distributions. We define a task descriptor vector $G \in \mathbb{R}^{d_G}$ whose components describe task-relevant object attributes and spatial relationships. Semantic masks provide a way to construct such quantities, including visibility and relative position. Let χ denote task-specified proprioceptive and execution context, selected in advance rather than identified with the unrestricted history. Suppose Z retains χ and supports an estimate $\widehat{G} \in \mathbb{R}^{d_G}$ of G. For $\rho \in \{\mu, \nu\}$, define $\delta_{G,\rho} = \mathbb{E}_\rho \|G - \widehat{G}\|$. These quantities must be defined under both distributions, with an explicit convention for unavailable components.

Assumption 5 (Descriptor-conditioned expert selection): There exists a common measurable rule g_{E} mapping descriptors and context to $\mathcal{A}$ such that, for $\rho \in \{\mu, \nu\}$,

$$\mathbb{E}_\rho \|\pi_{\mathrm{E}}(h) - g_{\mathrm{E}}(G, \chi)\| \le \epsilon_{\mathrm{sel},\rho}. \tag{12}$$

For a finite constant $K_{\mathrm{E}} \ge 0$,

$$\|g_{\mathrm{E}}(G, \chi) - g_{\mathrm{E}}(\widehat{G}, \chi)\| \le K_{\mathrm{E}} \|G - \widehat{G}\| \tag{13}$$

holds almost surely under both distributions, with finite $\epsilon_{\mathrm{sel},\rho}$ and $\delta_{G,\rho}$.

Here, $\epsilon_{\mathrm{sel},\rho}$ measures expert choices unexplained by (G, χ), while K_{E} controls sensitivity to estimation error. Task relevance alone does not imply this condition: equally successful actions may reflect different expert choices.

Define $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$. The triangle inequality yields

$$\mathbb{E}_\rho \|\pi_{\mathrm{E}}(h) - \widehat{a}_{\mathrm{E}}(Z)\| \le b_\rho := \epsilon_{\mathrm{sel},\rho} + K_{\mathrm{E}} \delta_{G,\rho}. \tag{14}$$

Appendix B-A gives the derivation. Supervising descriptor readouts targets the estimation term, while the choice of descriptors and context determines how well expert actions can be explained.

3) Bounds on Conditional Imitation-Error Discrepancy: Expert-action reconstruction connects information preservation to the discrepancy in Section III-B3. Because both $\pi(h) = p(Z)$ and $\widehat{a}_{\mathrm{E}}(Z)$ depend on the same representation, their squared difference is a common function under μ and ν. Expert reconstruction error bounds how far the actual conditional imitation errors depart from this function.

Assume a finite action-set diameter $B_{\mathcal{A}} = \sup_{a,a' \in \mathcal{A}} \|a - a'\|$ under a fixed action scale.

Proposition 1 (Conditional-error transfer through expert reconstruction): Under Assumption 5 and the representation-level coverage condition in Section III-B3, with b_ρ defined in (14),

$$D_\phi \le 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu). \tag{15}$$

If Assumption 4 also holds,

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C_\phi R_\nu(\pi) + 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu)}. \tag{16}$$

The proof is in Appendix B-B. The design objective is therefore to improve coverage while preserving information needed to reconstruct expert actions. For a fixed expert baseline, horizon, and action sensitivity, reducing the combined imitation and reconstruction terms tightens the task-failure bound.

D. Information-Conditioned Residual Corrections

Section III-C identifies conditions for preserving information needed to reconstruct expert actions. Such preservation does not eliminate the learned policy's imitation error, particularly when actions rely on outdated observations or limited execution history. We examine when additional observations and history make the remaining error more predictable, and when learned corrections realize this potential to reduce R_ν.

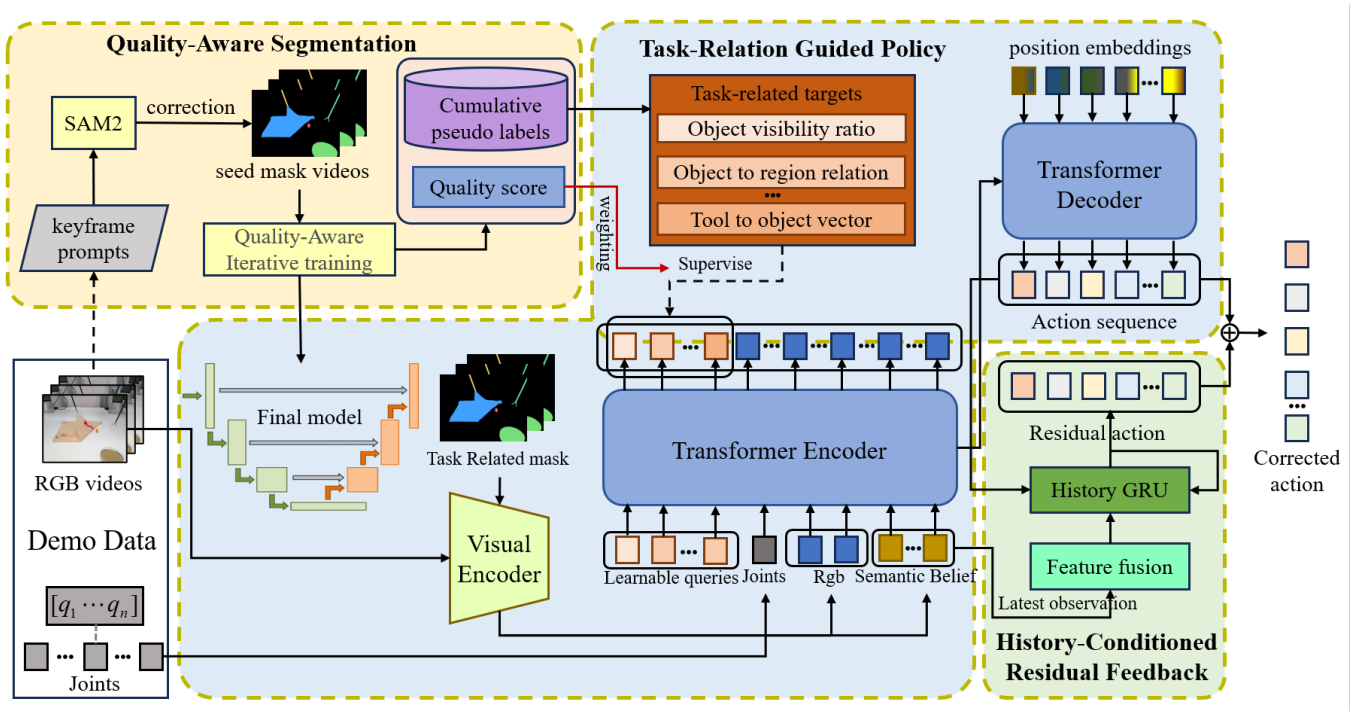


Fig. 1. Method overview. Quality-aware iterative annotation provides a frozen segmenter and quality-weighted descriptor labels. RGB, soft semantics, proprioception, and supervised queries jointly support action-chunk prediction. History-conditioned feedback uses updated observations and the current plan to correct unexecuted actions.

1) Residual Prediction of Action Error: We treat residual learning as prediction of the action deviation left by a fixed base policy. All comparisons use the same demonstration distribution ν, expert targets, target execution steps, and action scale. For base policy π_{b}, let $B = \pi_{\mathrm{b}}(\mathcal{H}_t)$ and define the residual target

$$E = \pi_{\mathrm{E}}(\mathcal{H}_t) - B. \tag{17}$$

Let $\mathcal{F}_{\mathrm{b}} \subseteq \mathcal{F}_{\mathrm{c}} \subseteq \mathcal{F}_{\mathrm{h}} \subseteq \sigma(t, \mathcal{H}_t)$ denote the information used to generate B, its augmentation with updated observations, and its further augmentation with execution history. The base action is $\mathcal{F}_{\mathrm{b}}$-measurable; all added information is available before the target action is executed. For any such information $\mathcal{F}$ and $\mathbb{E}_\nu\|E\|^2 < \infty$, the ideal residual predictor and its minimum squared error are

$$\begin{aligned} r^*_{\mathcal{F}} &= \mathbb{E}_\nu[E \mid \mathcal{F}], \\ R^*_\nu(\mathcal{F}) &= \mathbb{E}_\nu\|E - r^*_{\mathcal{F}}\|^2. \end{aligned} \tag{18}$$

This is an unrestricted square-integrable prediction benchmark, not an assumption that a network attains it or that $B + r^*_{\mathcal{F}}$ satisfies all action constraints. The derivation is in Appendix C-A.

2) Predictive Value of Additional Information: Additional information has strict predictive value when it changes the best estimate of the remaining action error. For $\mathcal{F}_0 \subseteq \mathcal{F}_1$,

$$R^*_\nu(\mathcal{F}_0) - R^*_\nu(\mathcal{F}_1) = \mathbb{E}_\nu\|r^*_{\mathcal{F}_1} - r^*_{\mathcal{F}_0}\|^2. \tag{19}$$

Applying this identity to $(\mathcal{F}_{\mathrm{b}}, \mathcal{F}_{\mathrm{c}})$ and $(\mathcal{F}_{\mathrm{c}}, \mathcal{F}_{\mathrm{h}})$ separates the potential contributions of updated observations and history. The gain is zero if the conditional mean is unchanged; more information alone does not imply a strict improvement. Appendix C-B gives the proof. These gains concern optimal prediction, not the performance of a particular history model.

3) Realized Corrections and Task-Performance Bounds: A learned correction improves imitation accuracy only when its prediction error is smaller than the error that the available information makes correctable. Let $r_{\mathcal{F}}$ be the square-integrable, $\mathcal{F}$-measurable residual actually applied, and let $\pi_{\mathcal{F}}(\mathcal{H}_t) = B + r_{\mathcal{F}}$. Define $\epsilon_{\mathrm{res}}(\mathcal{F}) = \mathbb{E}_\nu\|r_{\mathcal{F}} - r^*_{\mathcal{F}}\|^2$. Then

$$R_\nu(\pi_{\mathcal{F}}) = R_\nu(\pi_{\mathrm{b}}) - \mathbb{E}_\nu\|r^*_{\mathcal{F}}\|^2 + \epsilon_{\mathrm{res}}(\mathcal{F}). \tag{20}$$

Thus, R_ν decreases if $\epsilon_{\mathrm{res}}(\mathcal{F}) < \mathbb{E}_\nu\|r^*_{\mathcal{F}}\|^2$. The realization error includes model limitations, imperfect learning, and output constraints. Richer information improves the learned correction only if its predictive gain exceeds any increase in this error (Appendix C-C).

For admissible corrected actions, the task-performance implication follows from (11) under its coverage and action-sensitivity conditions. Corrections may change the deployment distribution, so the relevant quantity remains $C_\phi R_\nu(\pi) + D_\phi$ for each complete policy. For a fixed expert baseline, horizon, and positive action-sensitivity bound, reducing this combined quantity tightens the task-failure bound. This provides a conditional direction for improvement, not an ordering of actual success rates.

IV. Method

Section III motivates two directions: preserving information relevant to expert action selection and using new execution evidence to reduce remaining action error. Figure 1 organizes

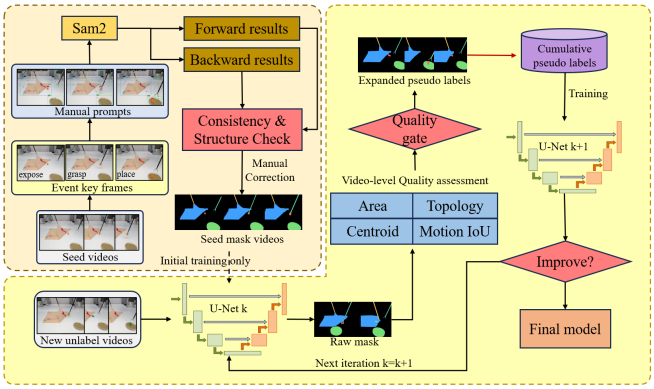


Fig. 2. Quality-aware iterative annotation. Reviewed bidirectional SAM2 masks train the initial segmenter. Subsequent models learn from cumulative, quality-screened pseudo-labels; seed videos are retained for reference rather than further training.

these directions into three components. First, quality-aware iterative annotation expands a small manually reviewed seed set into semantic supervision. Second, RGB observations, semantic maps, and supervised task-descriptor queries jointly support a base action-chunk policy. Third, a history-conditioned residual model uses updated observations to correct actions that have not yet been executed.

A. Quality-Aware Iterative Semantic Supervision

Task-descriptor labels are derived from semantic masks, so segmentation errors can compromise the supervision used to preserve task information. We combine reviewed seed annotations with video-level quality screening to limit error propagation as the labeled set expands (Fig. 2).

1) Event-Guided Seeds and Iterative Expansion: We select a small set of videos spanning the principal interaction and occlusion states, and annotate keyframes around events such as target exposure, contact, and placement. Bounding boxes and positive and negative point prompts guide bidirectional SAM2 propagation [23]; consistency checks, class-dependent structural processing, and manual correction produce the seed masks. The seeds train an initial U-Net segmenter, which predicts candidate masks on new unlabeled videos. Accepted masks accumulate across iterations, and each subsequent segmenter is reinitialized and trained on this cumulative pool without the original seed videos. The seed videos then serve as a fixed reference for monitoring segmentation accuracy and temporal anomalies. Iteration continues while anomaly rates improve without a material loss of segmentation accuracy.

2) Video-Level Quality Screening: The screening procedure distinguishes sustained task changes from transient segmentation anomalies. For each class, we extract trajectories of mask area, spatial extent, centroid, and topology, and fit robust piecewise trends over the video. Persistent changes can establish new segments, while short-lived departures provide anomaly evidence instead of penalizing every inter-frame change. Motion-compensated mask IoU supplies additional temporal evidence on pixels with forward–backward-consistent optical-flow correspondences. For class c at frame t, we collect these diagnostics as

$$\boldsymbol{\upsilon}_{t,c}^{\text{seg}} = (\upsilon_{t,c}^{\text{area}}, \upsilon_{t,c}^{\text{space}}, \upsilon_{t,c}^{\text{topo}}, \upsilon_{t,c}^{\text{motion}}). \tag{21}$$

Agreement across evidence groups identifies anomalous intervals; excessive changes introduced by structural processing also trigger exclusion. A frame enters subsequent segmentation training only when its relevant classes pass these checks. The manually reviewed reference set complements temporal screening by checking for persistent semantic errors that temporal consistency alone cannot detect. The same diagnostics also determine descriptor-specific supervision weights in Section IV-B. Quality assessment is performed offline; the final per-view segmenters are frozen and require only current images during policy execution.

B. Semantic Inputs and Task-Descriptor Queries

Semantic maps emphasize object-level spatial structure, while supervised queries encourage the joint representation to retain task-relevant attributes and relationships. RGB and proprioception remain available for decision cues not captured by the semantic classes. This design targets the descriptor-estimation term in (14) through the learned representation, rather than assuming that adding semantic channels improves coverage.

1) Joint Visual-Semantic Encoding: For view v, a frozen segmenter predicts class probabilities $p_t^v(c,\xi)$ from image I_t^v. We form a soft semantic image as

$$S_t^v(\xi) = \sum_{c \in \mathcal{C}_v} p_t^v(c,\xi) e_c^{\text{sem}}, \tag{22}$$

where $\mathcal{C}_v$ is the view-specific class set, ξ a pixel location, and $e_c^{\text{sem}} \in [0,1]^3$ a fixed class color. This probability-weighted color representation allows RGB and semantics to use the same three-channel visual encoder without argmax hardening. The encoded streams from all views are combined with proprioceptive and query tokens in a transformer encoder. An ACT decoder [6] reads this context to predict the base action chunk.

2) Supervised Descriptor Queries: We select K_{rel} descriptor groups $G^{(j)} \in \mathbb{R}^{d_j}$, such as visible area and relative position, which together form the task descriptor vector G. Fixed geometric operators applied to the offline masks generate pseudo-labels $\widetilde{G}_t^{(j)}$; the task-specific definitions are given in Appendix D-C. Each group has a learnable query whose encoded output $z_t^{(j)}$ supports a prediction head, $\widehat{G}_t^{(j)} = D_j(z_t^{(j)})$, and remains available to the action decoder. Supervision therefore encourages task information to be recoverable from the policy representation without prescribing an attention map.

A weight $\varpi_{n,j}^G \in [0,1]$ combines descriptor validity with quality evidence from the relevant semantic classes. Reliable labels receive full weight, uncertain labels are downweighted, and undefined or anomalous labels receive zero weight. In particular, a distance is undefined when either object is invisible, whereas a confirmed absence can provide a valid zero-visible-area label. Disabling a descriptor loss does not remove the

sample's action supervision. With descriptors normalized to fixed scales, the training objective is

$$\mathcal{L}_{\text{rel}} = \frac{1}{K_{\text{rel}}} \sum_{j} \frac{\sum_n \varpi_{n,j}^{G} \ell_G(\widehat{G}_n^{(j)} - \widetilde{G}_n^{(j)})}{\max(1, \sum_n \varpi_{n,j}^{G})},$$
$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{IL}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}}. \tag{23}$$

Here, n indexes minibatch samples, ℓ_G is the component-averaged Smooth L1 loss, and $\mathcal{L}_{\text{IL}}$ is the standard ACT training objective. The segmenters remain frozen, and quality weights affect auxiliary supervision rather than online policy inputs. Appendices D-A and D-B specify label admission and its connection to descriptor-estimation error.

C. History-Conditioned Residual Action Feedback

Even with task-relevant information, actions generated from an earlier observation may not match current execution needs. Following Section III-D, we freeze the base policy and train a residual model to predict its remaining action error from updated observations, plan context, and execution history.

1) Plan-Conditioned History Encoding: The base policy generates an action chunk and query context at time τ_p. A feedback update at τ_o predicts corrections for future execution times $t \geq \tau_o$. Its input record e_k^{fb} combines current visual and proprioceptive features, cached query features and target base actions, and recent command–response information. Query features are cached between plan updates, while images are re-encoded at each feedback update; relative timing identifies the actions being corrected. A GRU encodes the most recent L_{hist} records:

$$h_i^{\text{gru}} = \text{GRU}_\omega(e_i^{\text{fb}}, h_{i-1}^{\text{gru}}). \tag{24}$$

The hidden state is initialized to zero at the start of each window, and its final value h_k^{gru} is used for prediction. This history allows the model to interpret the current observation in light of previous commands and their observed responses.

2) Residual Prediction and Supervision: For each of H_r future slots, the residual head predicts a bounded adjustment to the corresponding base action:

$$r_{k,j} = \boldsymbol{\beta} \odot \tanh([W_r h_k^{\text{gru}} + b_r]_j),$$
$$\widehat{u}_{k,j} = B_{t_{k,j}} + r_{k,j}. \tag{25}$$

Here, $t_{k,j}$ is the target execution time, $B_{t_{k,j}}$ is the base command for that slot, and $\boldsymbol{\beta}$ bounds the residual in each action dimension.

Training replays the frozen base policy on demonstration trajectories, matching the observation and target-slot timing used for execution. For sample n and slot j, the residual target is $E_{nj} = u_{nj}^{\text{E}} - B_{nj}$, the demonstrated command minus the base command at the same execution step. Over valid sample–slot pairs $\mathcal{V}$, we minimize

$$\mathcal{L}_{\text{res}} = \frac{1}{|\mathcal{V}|} \sum_{(n,j)\in\mathcal{V}} \left[\ell_r\left(\frac{r_{nj} - E_{nj}}{\boldsymbol{\sigma}} \right) + \frac{\lambda_r}{d_u} \| \frac{r_{nj}}{\boldsymbol{\sigma}} \|^2 \right], \tag{26}$$

where ℓ_r is component-averaged Smooth L1, $\boldsymbol{\sigma}$ contains training-set action scales, and division is elementwise. Future demonstrated commands provide labels only and are not feedback inputs. This robust training objective is a surrogate for error reduction; the squared-error criterion in Section III-D is evaluated separately on held-out data. Replay learns corrections on demonstration histories, while their effect on subsequent execution is evaluated in closed-loop trials.

3) Execution Alignment: Each residual is attached to its originating plan and target slot; corrections for replaced plans or elapsed slots are discarded. The executed command is $u_t = B_t + \Lambda_t r_t$, where Λ_t indicates plan and timing validity, so an unavailable correction leaves the base command unchanged. This alignment applies the correction to the action it was trained to modify; the timing convention is summarized in Appendix D-E.

V. Experimental Evaluation

This section evaluates the proposed method on a bimanual exposure and execution task. The task requires coordinated actions under deformable occlusion, where preceding interactions change the visibility and accessibility of the target. We first assess overall task performance across object categories, initial positions, and occluder colors, then examine annotation reliability, semantic representations, and history-conditioned correction through controlled ablations. Finally, we evaluate the framework on a surgical-phantom task to assess its applicability beyond the primary setting.

The main experiments address four questions:

1) Does the complete method improve task success across objects, initial positions, and occluder colors?
2) Can quality-aware iteration reduce segmentation anomalies and annotation effort while preserving accuracy?
3) What do semantic inputs and task-descriptor supervision contribute, and how does perception quality affect their benefits?
4) Can recent observations and history reduce execution errors and support recovery from naturally occurring deviations?

A. Experimental Setup

1) Task and Platform: The platform consists of two SO-ARM101 arms, a paddle tool attached to the right arm, and fixed front and side cameras observing the tabletop workspace. Demonstrations involve three object categories: cubes, crumpled paper, and screws/nuts, initially covered by deformable cloth of one color. The arms first expose and separate the object from the cloth, push it into a designated region, and then restore cloth coverage in the front view.

Let $O_t^\star$, $R^\star$, and C_t denote the complete object projection, full target-region footprint, and cloth projection, respectively. The placement and restoration conditions are

$$p_{\text{in}}(t) = \mathbf{1}\{O_t^\star \subseteq R^\star\},$$
$$p_{\text{restore}}(t) = \mathbf{1}\{|C_t \cap \Omega| / |\Omega| \geq \theta_C\}, \tag{27}$$

where Ω is the full image domain of the fixed front camera and θ_C is the required cloth-coverage ratio. The task is complete when both conditions hold within the time limit:

$$\Phi(s_{0:T}) = \mathbf{1}\{\exists\, t \leq T : p_{\text{in}}(t) = 1 \wedge p_{\text{restore}}(t) = 1\}. \tag{28}$$

(Specify the time limit and cloth-coverage threshold.)

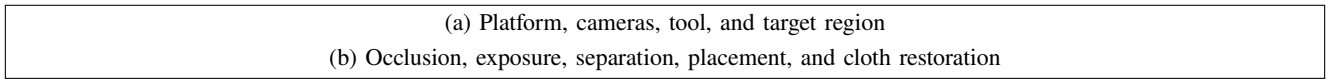


Fig. 3. Bimanual exposure and execution: the experimental platform and a complete task sequence.

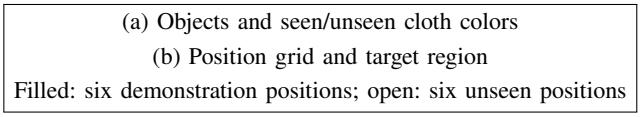


Fig. 4. Object categories, occluder colors, and initial positions. Filled and open markers indicate demonstration and unseen test positions, respectively.

TABLE I
Key training and execution settings.

Setting	Value
Data split / training seeds	TBD
Repetitions per condition	TBD
Time limit / coverage threshold	TBD
Control rate / deployment hardware	TBD
Chunk length / replanning interval	TBD
Residual interval / delay allowance	TBD

2) Demonstrations and Test Conditions: The dataset contains 90 demonstrations and 53,723 frame-aligned samples recorded at 30 Hz. Both camera streams have a resolution of 640×360. Proprioceptive states and actions are 10-dimensional joint-position vectors, with five joints per arm. Initial object positions follow a 3×4 grid, with demonstrations covering six positions (Fig. 4). The basic test set comprises all three demonstrated object categories at all twelve positions, yielding 36 conditions. Cloth of an unseen color tests appearance variation under matched object and position conditions. Each method receives the same number of independently reset trials per condition, with results reported separately for seen and unseen positions.

(Specify the unseen cloth color, test combinations, and repetitions per condition.)

3) Training Settings: All methods use a fixed demonstration budget, identical RGB and proprioceptive observations, the same action interface, and matched data splits. Semantic methods additionally use annotations derived from the demonstration videos. Base-policy training matches update counts and sample exposure; additional segmentation and residual-training resources are reported separately. Residual ablations share a frozen base policy, and model selection is separate from final testing. Table I summarizes execution settings; data usage and evaluation details appear in Appendices D-D and E.

4) Evaluation Metrics: The primary metric is full-task success within the time limit, reported as successes/total trials with uncertainty intervals. Additional metrics include completion time for successful trials, cumulative success over time using all trials as the denominator, stage attainment and conditional completion rates, and primary failure categories. Initial conditions and timing rules are matched across methods, with randomized execution order within condition blocks. Task success and stage progress are determined by independent video review.

B. Overall Task Performance

We compare overall performance and success across object categories, initial positions, and occluder colors. Baselines include ACT [6] and Diffusion Policy [8]; A2C2 [15] and VIOLA [12] are reserved as online chunk-correction and object-centric baselines, respectively.

(Complete baseline adaptation and tuning, and report additional resources. Distinguish ACT+A2C2 adaptation from both the original implementation and our MLP ablation.)

Model-selection and tuning budgets are matched, and inference speed is measured end to end on the deployment hardware.

1) Task Success and Completion Time: Table II reports success for the three demonstrated object categories, seen/unseen positions, and seen/unseen cloth colors, together with overall success and mean completion time for successful trials. Overall success counts each trial once in the predefined test pool; subgroup columns use their respective trial subsets. Figure 5 provides position-wise results.

(Compare overall success and time, then consistency across objects and unseen conditions. Use cumulative success to interpret timing; distinguish position shifts, color shifts, and their combination under the fixed demonstration budget.)

2) Completion Time, Stage Progress, and Failures: Figure 6 compares completion time and stage progress, while Table III summarizes primary failures. These results locate improvements in exposure, execution, or restoration; failure mechanisms are examined in the Discussion.

(Report success differences and intervals, locate stage-level gains, and state condition-specific limitations. Analyze causes using videos and logs in the Discussion; remove unimplemented optional baselines.)

C. Quality-Aware Semantic Annotation

We test whether quality-aware iteration reduces anomalies and the cost of reliable labels while preserving segmentation accuracy. Annotation development and independent review sets are separated by complete videos. We compare the seed model, iteration without quality filtering, and quality-filtered iteration using the same candidate pool and matched training budgets, recording the retained data volume. Independent review covers ordinary frames and exposure, contact, and placement events, reported separately rather than evaluated solely by the pipeline's quality scores.

Table IV reports raw mIoU, anomaly rate, label retention, and human effort. Anomalies are reviewed using predefined criteria for missing masks, incorrect assignments, and spurious structural changes. Temporal consistency uses motion-compensated IoU on valid optical-flow correspondences. Raw and postprocessed outputs are evaluated separately, with per-class object/tool IoU and anomaly counts aggregated by video.

TABLE II
Task success by object category, initial position, and occluder color. Success entries report count/total and percentage; mean completion time includes successful trials only.

Method	Object category			Position		Cloth color		Overall	
	Cube Seen	Paper Seen	Screws/nuts Seen	Seen 6 positions	Unseen 6 positions	Seen	Unseen	Success	Time (s)
ACT	–	–	–	–	–	–	–	–	–
Diffusion Policy	–	–	–	–	–	–	–	–	–
ACT+A2C2 (adapted)	–	–	–	–	–	–	–	–	–
VIOLA (adapted)	–	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–	–

Position-wise success: three object categories, ACT/DP/Ours, and cloth colors
3×4 grids with counts and demonstration positions; untested cells left blank

Fig. 5. Position-wise task success by object category and occluder color. Highlighted cells indicate demonstration positions; untested combinations remain blank.

(a) Cumulative task success over time
(b,c) Cumulative and conditional stage completion, with counts and intervals

Fig. 6. Completion-time and stage-level comparisons. All trials contribute to the denominator of the cumulative-success curve.

TABLE III
Primary failures, reported as counts and fractions of all trials for each method.

Method	Failure category	Count/total
TBD	Exposure incomplete	–
TBD	Ineffective contact	–
TBD	Object out of reach	–
TBD	Incomplete placement	–
TBD	Insufficient restoration	–
TBD	Other / undetermined	–

(a) Label volume, accuracy, and anomalies
(b) Quality traces and task events
(c) Raw masks, processed masks, and references

Fig. 7. Annotation expansion and temporal quality diagnostics, including genuine exposure and contact changes.

Human effort includes keyframe annotation, inspection, and correction, reported alongside the volume of labels meeting a common review standard. A manual frame-by-frame annotation/correction workflow on the same video subset provides a labor-cost reference.

(Compare filtering and iteration trends in accuracy, anomalies, retention, and effort. Separate raw-model improvements from postprocessing; claim time savings only when supported by recorded labor.)

(Verify annotation provenance before attributing existing labels to the latest pipeline. Report front/side results and all seed-annotation and review costs.)

D. Semantic Representation Ablations

Without residual feedback, we isolate semantic inputs, query capacity, and descriptor supervision. All groups use the same three objects and twelve positions with the demonstrated cloth color. We report success by object and seen/unseen position, stage completion, and failure categories; full condition-level results are provided as supplementary material. In Table V, S0–S3 progressively add semantics and queries. S2 and S3 share query count and architecture, differing only in auxiliary descriptor supervision. S3 and S4 use identical labels and segmenters but differ in quality weighting. S4 disables quality-based filtering and downweighting while retaining geometric-validity masks for undefined quantities. These base policies are trained separately with matched action supervision, views, and budgets. S5 fixes the S3 policy and replaces its inference-time segmenter with a lower-quality model selected on validation data. Class definitions, input resolution, and semantic encoding remain unchanged; latency differences are recorded separately. Training labels are unchanged, separating online perception degradation from changes in supervision quality.

Figure 8 summarizes task outcomes and stage-specific failures. On independently reviewed data, we additionally measure pseudo-label and deployment-mode descriptor errors for area, distance, and direction, together with label validity. An expert-action reconstruction probe examines whether descriptors retain action-selection information (Appendix E-E). These diagnostics examine the information-preservation pathway in Section III-C.

(Compare S1–S0, S3–S2, S3–S4, and S5–S3. Interpret class/position trends using stage outcomes, failures, and descriptor accuracy; these diagnostics do not directly estimate theoretical coverage constants.)

E. History-Conditioned Residual Feedback

Using one frozen RGB+semantics+QToken base policy, we test whether task context and history reduce execution errors and support natural recovery. Table VI crosses MLP/GRU

TABLE IV
Annotation quality and human effort on independently reviewed videos. The final iteration index k is not fixed in advance.

Model / labels	Videos	Raw mIoU	Raw anomaly rate	Motion IoU	Label retention	Human effort (min/video)
Seed model	–	–	–	–	–	–
Unfiltered iteration	–	–	–	–	–	–
Quality-filtered, round 1	–	–	–	–	–	–
Quality-filtered, round k	–	–	–	–	–	–

TABLE V
Representation ablations without residual feedback. UQ: queries without descriptor supervision; SQ: supervised task-descriptor queries.

ID	Inputs / queries	Quality weighting	Segmenter	Retrained	Success	Latency p95 (ms)
S0	RGB	N/A	None	Yes	–	–
S1	RGB + semantics	N/A	Standard	Yes	–	–
S2	RGB + semantics + UQ	N/A	Standard	Yes	–	–
S3	RGB + semantics + SQ	On	Standard	Yes	–	–
S4	RGB + semantics + SQ	Off	Standard	Yes	–	–
S5	Same as S3	Same as S3	Lower quality	No	–	–

(a) Success by object and position; (b) Stage completion; (c) Failure categories
(d) Descriptor errors and valid-label fractions, with sample counts and intervals

Fig. 8. Representation ablations: task success, stage progress, failure categories, and descriptor accuracy.

correction with access to semantic and QToken features, which are supplied jointly.

1) Matched Inputs and Residual Models: All residual models receive the latest RGB observation, proprioception, the current base plan, and target-slot information. Models without semantic context also exclude Transformer features fused with semantics or QTokens, while retaining the same semantic base actions. Context-enabled models additionally receive semantic features and cached QTokens, with their respective observation timestamps recorded.

(Fix whether residual semantic features are fresh or cached, use the same convention across groups, and report segmentation latency.)

The MLP uses the current record; the GRU uses a history window containing that same record. Residual training matches action targets, update intervals, output horizons, clipping, and data splits. Inputs such as velocity or previous commands are specified explicitly, and a capacity-matched current-information model checks whether gains arise merely from parameter count.

F0 shares the S3 base policy, and F4 is the complete method. A2C2 is adapted to the same frozen base for an external correction-method comparison, matching available inputs and training budgets while measuring execution-latency differences. RGB ACT with RGB-history correction provides a connecting comparison with S0, S3, and F4. More frequent replanning and temporal ensembling on the same base policy provide execution-scheme controls, with actual compute and update rates reported.

2) Error Reduction and Natural Recovery: Groups use the object and position conditions of the representation ablations without artificial disturbances. Alongside subgroup success and stage outcomes, full execution videos are reviewed for cloth entanglement, ineffective contact, transport deviation, and execution stagnation. These events may occur in successful or failed trials; local recovery and eventual task completion are recorded separately. Error incidence is the fraction of trials containing at least one error. Recovery is the fraction of identified events resolved within a predefined window, supplemented by recovery time and final success among error-containing trials. Recovery is not applicable when no events occur. Table VII breaks down results by error type; event and counting rules appear in Appendix E-D. Because encountered errors can differ across policies, recovery statistics are interpreted jointly with overall success and error incidence.

3) Correction Accuracy and Runtime: Offline correction is evaluated on matched replay samples, expert targets, execution slots, and action scales, reporting base/corrected MSE and the alignment benefit and correction energy in (43). Table VIII records end-to-end latency, observation age, effective update rate, and expired corrections. Candidate corrections and executed commands are logged separately. A control retaining feedback computation but suppressing its output can additionally check scheduling effects.

(Compare semantic-context and history contrasts, then link matched-slot error, timely execution, error incidence, recovery, and final success. Compare A2C2 latency and correction quality; include harmful corrections and failed recovery.)

F. Cross-Task Applicability

We further evaluate the framework beyond tabletop pushing by training each method on surgical-phantom demonstrations.

TABLE VI
Residual ablations and a correction-method comparison with one frozen base policy. Error incidence and recovery concern naturally occurring execution errors.

ID	Model	Semantic + QToken input	History	Success	Error incidence	Recovery	Action MSE
F0	None	N/A	N/A	–	–	–	–
F1	MLP	No	No	–	–	–	–
F2	MLP	Yes	No	–	–	–	–
F3	GRU	No	Yes	–	–	–	–
F4	GRU	Yes	Yes	–	–	–	–
Baseline	A2C2 (adapted)	Matched	Original	–	–	–	–

TABLE VII
Natural-error and recovery diagnostics, instantiated for each method. Final success uses error-containing trials; recovery time includes recovered events only.

Method	Error type	Error trials / all trials	Recovered / events	Final success after error	Recovery time (s)
TBD	Cloth entanglement	–	–	–	–
TBD	Ineffective contact	–	–	–	–
TBD	Transport deviation	–	–	–	–
TBD	Execution stagnation	–	–	–	–
TBD	Deduplicated total	–	–	–	–

TABLE VIII
Correction accuracy and runtime. Offline errors use matched samples; runtime is measured on the deployment device.

Corrector	Alignment benefit	Correction energy	Latency p50/p95 (ms)	Observation age (ms)	Update rate (Hz)	Expired fraction
F1	–	–	–	–	–	–
F2	–	–	–	–	–	–
F3	–	–	–	–	–	–
F4	–	–	–	–	–	–
A2C2 (adapted)	–	–	–	–	–	–

(a) Subgroup success and stages; (b) Errors, recovery, and final outcomes
(c) Error–correction–recovery sequences and synchronized commands, including failures

Fig. 9. Error reduction and natural recovery with residual feedback. Mark error onset, recovery, and plan/feedback updates in synchronized traces.

(a) Phantom setup, tools, and cameras
(b) Occlusion, retraction, grasping, and placement

Fig. 10. Surgical-phantom exposure and execution: the platform and a complete task sequence.

The semantic supervision, task-descriptor queries, and history-conditioned residual design are retained to assess cross-task applicability rather than zero-shot policy transfer.

1) Surgical-Phantom Task and Adaptation: The arms retract tissue-mimicking material, expose an occluded object, grasp it, and transfer it into a designated region. Unlike pushing, this task requires grasping contact coordinated with exposure. The provisional completion condition requires full placement and release, with exposure, stable grasping, and placement as intermediate milestones.

(Finalize completion and restoration requirements, timing, hardware, action interface, dataset, test conditions, repetitions, and safety-stop rules.)

All methods use the same task-specific demonstrations and evaluation conditions, with model selection separate from testing. Adaptations to the action interface, semantic classes, and task descriptors are reported alongside shared training and architectural choices.

(List retained and adapted components, annotation effort, and training cost; explain changes to descriptors, camera placement, or control rate.)

2) Overall Comparison and Key Ablations: Table IX compares ACT, DP, and the complete method, retaining semantic-input, semantic-query, and MLP-residual controls. These comparisons test whether representation and history benefits persist without repeating the full ablation suite. Residual variants share the same semantic-query base trained on the surgical-phantom data.

(Compare overall and stage-level gains with the primary task. Merge placement and full success if their criteria coincide; limit conclusions to evaluated applications, not zero-shot transfer or clinical effectiveness.)

TABLE IX
Surgical-phantom results. E/G/P: exposure, grasping, and placement attainment. Sem./Q: semantics and QToken. Time includes successful trials only.

Method	Success	Time (s)	E	G	P
ACT	–	–	–	–	–
Diffusion Policy	–	–	–	–	–
ACT + Sem.	–	–	–	–	–
ACT + Sem. + Q	–	–	–	–	–
ACT + Sem. + Q + MLP	–	–	–	–	–
Ours (GRU residual)	–	–	–	–	–

VI. Discussion and Conclusion

A. Failure Cases and Limitations

本节结合表 III的失败分布和同步视频、指令、感知日志，分析完整方法尚未解决的问题。案例选择覆盖不同失败类型，也包含残差未能恢复及修正后进展倒退的情况，不以个别成功恢复片段替代整体统计。

1) Perception and Task Information: 质量筛选针对视频中的异常变化，不能排除时间上稳定的系统性误分。所选描述量刻画对象属性与空间关系，但不一定足以解释接触状态或专家的具体动作选择；语义分割准确与描述量读出准确也不意味着策略已经正确利用这些信息。

（结合图 11(a,b) 区分分割错误、描述量估计错误和执行决策错误。说明是否存在描述量准确仍失败的情况，并联系理论中的专家选择近似误差解释未被当前信息覆盖的动作选择因素。没有探针或干预证据时，将原因表述为待验证解释，而非确定归因。）

2) Residual Prediction and Execution: 历史提供新的动作–响应证据，但有限数据、修正幅度和执行延迟可能限制其作用。依据第 III-D节，实际收益取决于可预测误差与残差实现误差的关系，而不是仅由是否使用 GRU 决定。

（结合图 11(c) 和表 VII区分预测不准、时效不足、修正过大和超出可恢复范围的案例；比较 MLP 与 GRU 在哪些条件下出现不同结果。修正之后发生失败仅说明时间关联，若无匹配对照，不直接声称由残差造成。）

3) Scope of the Evidence: 本文的任务表现界使用总体模仿误差，并依赖覆盖、动作敏感性和专家选择条件。独立轨迹上的读出和动作误差用于检验相应机制，不能将有限训练损失直接代入这些界，也不能据更紧的上界确定实际成功率排序。两个任务上的分别训练验证方法的适用性，不等同于零样本跨任务迁移；手术模拟材料上的实验亦不构成临床有效性证据。

（根据真实划分与训练重复说明结论覆盖哪些物体、位置和平台。联系理论 B–D 总结哪些机制获得实验支持、哪些条件尚未得到直接测量；概括新增任务的描述量设计、语义标注与训练成本。若未做多预算实验，不声明样本效率优势；QToken 没有独立停止头时，不将任务成功等同于自主停止能力。）

B. Conclusion

（用一段总结问题、三个方法模块及两个实机任务支持的核心结论。填写实际成功率、稳定性或恢复收益，只保留与匹配对照相符的表述；最后指出一个由失败分析直接支持的主要局限，不预先声明显著提升、通用性或收敛保证。）

Appendix A

A. Proof of the Policy Performance-Difference Identity

We prove (8) by summing expected changes in the reference value. The evaluation-domain restriction in Section III-A and Assumption 1 give the terminal value

$$V_T^{\pi_0}(\mathcal{H}_T) = 1 - \mathcal{V}(I_{0:T}^{1:V}) = 1 - \Psi. \tag{29}$$

For $t < T$, taking action $\pi(\mathcal{H}_t)$ and then following π_0 gives the failure probability $V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})$ conditional on the next history. Averaging over next histories under Assumption 2 gives

$$Q_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = \mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]. \tag{30}$$

Here, π determines the next-history distribution, while $V_{t+1}^{\pi_0}$ accounts for subsequent execution under π_0.

Combining (7) and (30) and taking expectations gives

$$\begin{aligned}&\mathbb{E}_\pi[A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))]\\ &= \mathbb{E}_\pi\left[\mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]\right] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)].\end{aligned} \tag{31a}$$

By the tower property, averaging the conditional next-step value over current histories gives its unconditional expectation:

$$\begin{aligned}&\mathbb{E}_\pi\left[\mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]\right]\\ &= \mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})].\end{aligned} \tag{31b}$$

Substituting (31b) into (31a) and summing over time leaves only the terminal and initial terms:

$$\begin{aligned}&\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))\\ &= \sum_{t=0}^{T-1}\left\{\mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)]\right\}\\ &= \mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] - \mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)].\end{aligned} \tag{31c}$$

At time T, no action remains. By (29), the terminal term is the failure probability under π:

$$\mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] = \mathbb{E}_\pi[1 - \Psi] = F(\pi). \tag{31d}$$

The policies share the initial-history distribution, so the initial term equals the failure probability of following π_0 throughout:

$$\begin{aligned}\mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)] &= \mathbb{E}_{\pi_0}[V_0^{\pi_0}(\mathcal{H}_0)]\\ &= F(\pi_0).\end{aligned} \tag{31e}$$

Finally, substituting (31d) and (31e) into (31c) gives

$$\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = F(\pi) - F(\pi_0). \tag{31f}$$

This establishes (8).

(a) 持续分割错误：RGB、语义预测、人工审核
(b) 描述量准确但执行失败：任务关系与接触状态
(c) 未恢复或进展倒退：基础计划、残差、实际运动与时间戳
填写两个任务的代表性案例；只标注日志与视频支持的事实

Fig. 11. Representative failure cases and supporting evidence. Placeholder for perception, task-information, and residual-execution limitations.

B. Proof of the Task-Failure Bound

We prove (9) by first bounding task consequences through action deviations and then transferring the resulting deployment error to the demonstration distribution.

Set $\pi_0 = \pi_{\mathrm{E}}$ and let $e(h) = \|\pi(h) - \pi_{\mathrm{E}}(h)\|$. Applying Assumption 4 in (8) bounds each step's increase in failure probability by its action deviation:

$$F(\pi) - F(\pi_{\mathrm{E}}) \le L\,\mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} e(\mathcal{H}_t)\right]. \tag{32a}$$

Since $\bar{\mu}_\pi$ samples control steps uniformly, the time sum equals T times the deployment average:

$$\mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} e(\mathcal{H}_t)\right] = T\,\mathbb{E}_{(t,h)\sim\bar{\mu}_\pi} e(h). \tag{32b}$$

By the Cauchy–Schwarz inequality, the mean action deviation is bounded by its root mean square:

$$\mathbb{E}_{\bar{\mu}_\pi} e(h) \le \sqrt{\mathbb{E}_{\bar{\mu}_\pi} e(h)^2}. \tag{32c}$$

Assumption 3 also bounds expectations of nonnegative measurable functions, by simple-function approximation and monotone convergence. Applying it to $e(h)^2$ and using (3) gives

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \le C\,\mathbb{E}_{\nu} e(h)^2 = C R_\nu(\pi). \tag{32d}$$

Combining (32a)–(32d) establishes (9).

C. Proof of the Representation-Level Transfer Bound

We prove (10) by separating conditional error discrepancy from distributional coverage, then apply action sensitivity to obtain (11).

Let $\mu = \bar{\mu}_\pi$, with the remaining notation and the integrability and measurable conditional-mean conditions as in Section III-B3. Representation coverage ensures that ν_Z-null sets are also μ_Z-null sets, so the demonstration-side conditional mean is well defined for the deployment comparison.

First, conditioning on the representation gives

$$\mathbb{E}_{\mu} e(h)^2 = \mathbb{E}_{\mu_Z} m_\mu(Z). \tag{33a}$$

At the same representation, deployment and demonstration conditional errors can differ. By the definition of D_ϕ,

$$\begin{aligned}\mathbb{E}_{\mu_Z} m_\mu(Z) &\le \mathbb{E}_{\mu_Z} m_\nu(Z) + \mathbb{E}_{\mu_Z}|m_\mu(Z) - m_\nu(Z)| \\ &= \mathbb{E}_{\mu_Z} m_\nu(Z) + D_\phi.\end{aligned} \tag{33b}$$

Finally, applying representation coverage to the nonnegative function m_ν and taking the total expectation give

$$\mathbb{E}_{\mu_Z} m_\nu(Z) \le C_\phi \mathbb{E}_{\nu_Z} m_\nu(Z) = C_\phi R_\nu(\pi). \tag{33c}$$

Combining (33a)–(33c) establishes (10). Under Assumption 4, substituting this result into the sensitivity, time-average, and Cauchy–Schwarz steps (32a)–(32c) yields (11).

Appendix B
Proofs for Task-Relevant Information Preservation

A. Proof of the Expert-Action Reconstruction Bound

We prove (14) under the common encoder and readouts of Section III-C, with $\mu = \bar{\mu}_\pi$. Define the common represented expert rule $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$. The triangle inequality and Assumption 5 give, for $\rho \in \{\mu, \nu\}$,

$$\begin{aligned}\mathbb{E}_\rho\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| &\le \mathbb{E}_\rho\|\pi_{\mathrm{E}} - g_{\mathrm{E}}(G,\chi)\| \\ &\quad + \mathbb{E}_\rho\|g_{\mathrm{E}}(G,\chi) - g_{\mathrm{E}}(\widehat{G},\chi)\| \\ &\le \epsilon_{\mathrm{sel},\rho} + K_{\mathrm{E}}\delta_{G,\rho} = b_\rho.\end{aligned} \tag{34}$$

B. Proof of the Conditional-Error Transfer Bound

We prove Proposition 1 by comparing actual imitation error with a common function of Z, then transferring the demonstration-side reconstruction error through representation coverage. Use $\pi(h) = p(Z)$ and $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$ from Section III-C. Since $\pi(h) = p(\phi(t,h))$, the represented squared error $\ell_Z(z) = \|p(z) - \widehat{a}_{\mathrm{E}}(z)\|^2$ is the same function of z under both distributions. All three actions belong to $\mathcal{A}$, so the reverse triangle inequality yields

$$\begin{aligned}|e(h)^2 - \ell_Z(Z)| &\le \big(\|\pi - \pi_{\mathrm{E}}\| + \|\pi - \widehat{a}_{\mathrm{E}}\|\big)\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| \\ &\le 2B_{\mathcal{A}}\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\|.\end{aligned} \tag{35}$$

Conditioning separately under each distribution gives

$$|m_\rho(z) - \ell_Z(z)| \le 2B_{\mathcal{A}}\mathbb{E}_\rho[\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| \mid Z = z]. \tag{36}$$

Insert the common function ℓ_Z between m_μ and m_ν and apply representation coverage to the nonnegative demonstration-side conditional error:

$$\begin{aligned}D_\phi &\le \mathbb{E}_{\mu_Z}|m_\mu(Z) - \ell_Z(Z)| \\ &\quad + \mathbb{E}_{\mu_Z}|m_\nu(Z) - \ell_Z(Z)| \\ &\le \mathbb{E}_{\mu_Z}|m_\mu(Z) - \ell_Z(Z)| \\ &\quad + C_\phi\mathbb{E}_{\nu_Z}|m_\nu(Z) - \ell_Z(Z)| \\ &\le 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu).\end{aligned} \tag{37}$$

Representation coverage ensures that the demonstration-side conditional inequality also holds μ_Z-almost everywhere. This proves (15); substitution into (11) establishes (16).

Appendix C
Proofs for Information-Conditioned Residual Corrections

All proofs fix the base policy, expert targets, and distribution ν from Section III-D.

A. Proof of the Optimal Residual Predictor

For any square-integrable $\mathcal{F}$-measurable vector v, the tower property gives

$$\begin{aligned}\mathbb{E}_\nu\langle E - r^*_{\mathcal{F}}, v\rangle &= \mathbb{E}_\nu\langle \mathbb{E}_\nu[E - r^*_{\mathcal{F}} \mid \mathcal{F}], v\rangle \\ &= 0.\end{aligned} \tag{38}$$

For any square-integrable $\mathcal{F}$-measurable predictor r, write $E - r = (E - r^*_{\mathcal{F}}) + (r^*_{\mathcal{F}} - r)$. Expanding the square and applying (38) yields

$$\mathbb{E}_\nu\|E - r\|^2 = R^*_\nu(\mathcal{F}) + \mathbb{E}_\nu\|r - r^*_{\mathcal{F}}\|^2. \tag{39}$$

The second term is nonnegative and vanishes at $r = r^*_{\mathcal{F}}$, proving (18).

B. Proof of the Information-Gain Identity

For $\mathcal{F}_0 \subseteq \mathcal{F}_1$, the tower property gives $\mathbb{E}_\nu[r^*_{\mathcal{F}_1} \mid \mathcal{F}_0] = r^*_{\mathcal{F}_0}$. Since $r^*_{\mathcal{F}_0}$ is also $\mathcal{F}_1$-measurable, applying (39) with information $\mathcal{F}_1$ and predictor $r^*_{\mathcal{F}_0}$ gives

$$R^*_\nu(\mathcal{F}_0) = R^*_\nu(\mathcal{F}_1) + \mathbb{E}_\nu\|r^*_{\mathcal{F}_1} - r^*_{\mathcal{F}_0}\|^2. \tag{40}$$

Rearranging proves (19). The gain is strictly positive exactly when the two conditional means differ on a set of positive ν-probability.

C. Proof of the Realized-Correction Criterion

Setting $r = 0$ in (39) gives

$$R_\nu(\pi_{\mathrm{b}}) = R^*_\nu(\mathcal{F}) + \mathbb{E}_\nu\|r^*_{\mathcal{F}}\|^2. \tag{41}$$

Setting $r = r_{\mathcal{F}}$ in the same identity gives $R_\nu(\pi_{\mathcal{F}}) = R^*_\nu(\mathcal{F}) + \epsilon_{\mathrm{res}}(\mathcal{F})$. Combining these expressions proves (20). For two learned corrections under nested information, subtraction further gives

$$\begin{aligned}R_\nu(\pi_{\mathcal{F}_0}) - R_\nu(\pi_{\mathcal{F}_1}) = \mathbb{E}_\nu\|r^*_{\mathcal{F}_1} - r^*_{\mathcal{F}_0}\|^2 \\ + \epsilon_{\mathrm{res}}(\mathcal{F}_0) - \epsilon_{\mathrm{res}}(\mathcal{F}_1).\end{aligned} \tag{42}$$

Thus, the predictive gain must exceed any additional realization error for richer information to improve the learned policy.

Equivalently, direct expansion gives an error diagnostic without estimating the ideal predictor:

$$R_\nu(\pi_{\mathrm{b}}) - R_\nu(\pi_{\mathcal{F}}) = 2\mathbb{E}_\nu\langle E, r_{\mathcal{F}}\rangle - \mathbb{E}_\nu\|r_{\mathcal{F}}\|^2. \tag{43}$$

Here $r_{\mathcal{F}}$ includes output processing and is zero when no correction is applied; it must remain measurable with respect to the available information. For admissible actions, apply (11) to each complete policy using its own deployment distribution and representation. A reduction in R_ν tightens that bound when the other terms are held fixed, or more generally when $C_\phi R_\nu + D_\phi$ decreases with a common expert baseline, horizon, and positive sensitivity bound. The identities concern population error under ν; replacing it with training loss requires a separate generalization argument.

Appendix D
Method Details

A. Descriptor Labels and Quality Weights

Descriptor supervision uses the quality evidence relevant to its geometric definition rather than a single frame-wide score. Visible area depends primarily on target extent and topology; relative position also depends on the localization of both objects, and a tool-endpoint vector is sensitive to tool shape. For the classes $\mathcal{C}_j$ used by descriptor j, let $\eta^{\mathrm{seg}}_{t,c}$ indicate that class c passes anomaly screening, and let $m^G_{t,j}$ indicate that the descriptor is defined. The supervision weight is

$$\varpi^G_{t,j} = m^G_{t,j} \prod_{c \in \mathcal{C}_j} \eta^{\mathrm{seg}}_{t,c}\, g_j(\upsilon^G_{t,j}), \tag{44}$$

where $\upsilon^G_{t,j}$ aggregates the relevant evidence in (21), and g_j is a nonincreasing weight between zero and one. Reviewed descriptor labels determine the reliable, downweighted, and rejected ranges; these weights are not probabilities of correctness. Without sufficient evidence to calibrate a graded weight, $g_j \equiv 1$ gives validity and anomaly admission alone. The groupwise normalization in (23) primarily changes the relative contributions of samples within a descriptor group.

An empty mask is a valid zero-area label only when absence is confirmed by review or supporting temporal evidence. Undefined distances and directions are excluded from the auxiliary loss rather than labeled as zero. Missing optical-flow correspondences provide no motion evidence, whereas excessive structural repair excludes a frame from segmentation training. Whole-video diagnostics may use later frames offline; they are not supplied to the deployed policy.

B. Pseudo-Label and Descriptor-Readout Errors

For either evaluation distribution ρ used in Section III-C, inserting the descriptor pseudo-label $\widetilde{G}$ gives

$$\delta_{G,\rho} \le \mathbb{E}_\rho\|\widehat{G} - \widetilde{G}\| + \mathbb{E}_\rho\|\widetilde{G} - G\|. \tag{45}$$

The terms separate descriptor fitting from pseudo-label error, motivating quality screening in addition to query supervision. When measured only on admitted labels, this comparison concerns the admitted subset; the admission rate is reported alongside the error.

C. Task-Descriptor Instance

Our task instance uses front-view masks for three descriptor groups, irrespective of whether the policy receives one or two camera views. They describe target visibility, target–region distance, and the vector from a tool endpoint to the target. For image width W and height H_I, let M_O, M_R, M_T be the object, region, and tool masks, with centroids c_O, c_R, c_T when nonempty. The labels are

$$\begin{aligned}\widetilde{G}^{(1)} &= |M_O|/(WH_I), \\ \widetilde{G}^{(2)} &= \|c_O - c_R\|/\sqrt{(W-1)^2 + (H_I-1)^2}, \\ \widetilde{G}^{(3)} &= \mathrm{clip}(\bar{c}_O - \bar{e}_{T,L}, -1, 1).\end{aligned} \tag{46}$$

Here, $\bar{c} = (c_x/(W-1), c_y/(H_I-1))$, with rightward and downward positive directions. The endpoint $\bar{e}_{T,L}$ is the leftmost of the two extremes along the tool mask's principal axis in normalized coordinates. This geometric convention need not identify the physical contact tip. Target–region centroid distance is an auxiliary descriptor, not the criterion for complete target containment.

D. Evaluation Scope

Seed videos used by the initial segmenter provide a development reference, not an independent final test. Cross-video evaluation requires held-out videos excluded from annotation-model fitting and selection, with the split maintained through policy and residual training. For existing checkpoints, the documented residual train–validation split establishes only residual-head validation; it does not establish that upstream models were trained on disjoint episodes. Descriptor evaluation and residual-cache generation use the base policy in deployment mode. For ACT, this sets the latent input to zero rather than conditioning on the demonstrated action posterior [6].

E. Residual Targets and Execution Alignment

Residual targets use the same demonstrated command stream and action scale as the base policy; measured joint states remain observations. Training normalizes errors by per-dimension action scales, while residuals are added to the base command in matching command units. On the discrete control grid, a feedback observation at step n_k with reserved delay d_{fb} targets

$$t_{k,j} = n_k + d_{\text{fb}} + j, \qquad j = 0, \ldots, H_r - 1. \tag{47}$$

Online timestamps determine whether each target slot is still available; a late correction is not shifted onto a different action. Plan replacement invalidates residuals attached to the previous plan. Timing evaluation measures observation-to-execution latency, including acquisition, encoding, and scheduling, rather than network inference alone.

(定稿前须完成统一数据划分及新版描述量质量权重的实现核对；已有结果不能自动作为新版方法的验证。待核对事项单列于 method_implementation_notes.md。)

Appendix E
Evaluation Details

A. Data and Input Details

Front-view annotations contain cloth, object, target region, tool, left arm, and right arm; side-view annotations contain the first four classes. Both views also include background.

(Specify action targets, normalization, and image preprocessing from the final checkpoints. Verify actual splits and model selection: dataset metadata lists all 90 episodes as training data. Split validation by complete episodes and distinguish acquisition, control, and inference rates.)

TABLE X
Evaluation milestones for bimanual exposure and execution.

Milestone	Definition
Exposure	The object becomes identifiable and meets the visibility threshold.
Separation	Object and cloth boundaries are nonadjacent, with sufficient visibility for assessment.
Placement	The entire object lies within the independently defined target region.
Restoration	Cloth image coverage reaches the threshold while the object remains fully inside the region.

B. Visual Completion and Milestone Review

The front camera is the primary evaluation view; the side camera assists review of occlusion and ambiguous boundaries. The full target region is defined by an independent reference contour rather than its partially visible segmentation. An empty object mask is not evidence of complete placement, and unobservable intervals are not counted as confirmed success. The object must remain inside the region at completion. Table X defines evaluation milestones without requiring an explicit phase-prediction head.

Let N_j be the number of trials completing the first j milestones in order and N the total number of trials. Cumulative and conditional completion rates are N_j/N and N_j/N_{j-1}, respectively, with $N_0 = N$. Zero denominators are reported as not applicable, and repeated stage entries are not counted again. All methods share the same timing origin. Completion time is the first instant satisfying the full criterion; if persistence is required, use the end of confirmation and also record initial satisfaction.

(Specify reference contours, boundary tolerance, visibility and adjacency thresholds, confirmation duration, reviewers, disagreement resolution, and unassessable trials. Reflect any allowed placement tolerance in the main criterion.)

C. Trial Allocation and Accounting

Demonstration positions are $(0,0)$, $(0,3)$, $(1,1)$, $(1,2)$, $(2,0)$, and $(2,3)$. Basic tests cover the three demonstrated objects at all twelve positions; extended tests use an unseen cloth color. Methods share object–position–color combinations and reset rules. Each trial records the object, position, cloth color, checkpoint, training seed, repetition, completion time, and primary failure. Overall and subgroup results use the same trial pool with explicit denominators. Overall success counts each trial once, rather than summing overlapping subgroup columns. Position and color comparisons match the remaining conditions and distinguish individual from combined shifts. Policies and segmenters are frozen before testing, and segmentation-development exposure to the nominally unseen conditions is checked.

(Specify unseen colors, test combinations, and subgroup denominators. Disclose conditions seen during segmentation development; keep cloth material, dimensions, and thickness matched when changing color.)

Key comparisons use independent training seeds, with intervals respecting seed and condition-block structure. Repeated executions of one checkpoint do not substitute for independent training runs. Each trial receives one primary failure category; other process anomalies may be recorded separately.

(Verify episode-to-condition mappings, repetitions, seeds, interval estimation, grid spacing, and reset rules. Basic tests contain 36 times the repetitions per condition; count extended tests separately. Prespecify timeouts, safety stops, technical failures, and retry accounting.)

D. Natural Error and Recovery Review

Natural errors are identified from all execution videos and logs, not only successful recovery clips. The following observable criteria are fixed before evaluation:

- *Cloth entanglement:* cloth obstructs tool or object motion; recovery requires clearing the obstruction and resuming task progress.
- *Ineffective contact:* an attempted push fails to establish effective contact or object progress; recovery requires subsequent effective contact and displacement.
- *Transport deviation:* the object leaves the transport path or remains at the region boundary; recovery requires renewed transport toward the region, with full placement recorded separately.
- *Execution stagnation:* no effective progress occurs for a predefined duration before completion; recovery requires renewed progress in the current stage.

An event begins when its criterion is first met and ends at recovery, task termination, or expiration of the recovery window. Recovery requires a confirmation duration; termination without recovery counts as unresolved. Continuous events are counted once; a new event requires recovery followed by recurrence. Overlapping phenomena receive one primary category and optional auxiliary labels, rather than being split into multiple recovery opportunities. Error incidence counts trials with at least one event. Recovery time includes recovered events only, alongside recovered/total events and final successes/error-containing trials. Recovery is not applicable when no events occur; unassessable events are reported separately.

(Specify progress criteria, tolerances, stagnation and recovery durations, confirmation, and category precedence. Report severity and counts by condition, use trial-clustered intervals, and list events with incomplete follow-up. Include failed recovery and harmful correction examples.)

E. Representation Diagnostics

Expert-action reconstruction probes compare context alone, reviewed descriptors with context, and predicted descriptors with context. Readout capacity and action targets are matched, with training and evaluation separated by complete trajectories. Context is identical across probes and excludes target actions and future observations. Evaluation reports descriptor error, valid-label fractions, and action-reconstruction error. These probes examine the relation between task information and action selection; they neither replace deployment-side expert labels nor directly estimate coverage or conditional error discrepancy.

(Specify context, action scales, readout models, and independent trajectories. Document the causal sources and time ranges of velocity or previous-command inputs.)

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